Author Rebuttal for Submission ICML 2026

Authors

MIM Attack

Table 1: Mean Robustness Ratios ($Adv/Clean$) $\pm$ std for the MIM Attack across budgets ϵ. Architecture families: **ShuffleNetV2** (SN) and **RepVGG** (RV). r denotes the Pearson correlation coefficient.

Fam.	Model	$N_{eff} \downarrow$	$\epsilon = \frac{1}{255}$	$\epsilon = \frac{2}{255}$	$\epsilon = \frac{4}{255}$	$\epsilon = \frac{8}{255}$	$\epsilon = \frac{16}{255}$
			CIFAR-10				
RV	`repvgg_a0`	78.1 ± 0.12	0.522 ± 0.006	0.451 ± 0.007	0.351 ± 0.006	0.238 ± 0.005	0.206 ± 0.005
	`repvgg_a1`	78.6 ± 0.15	0.521 ± 0.005	0.450 ± 0.007	0.347 ± 0.008	0.237 ± 0.009	0.206 ± 0.007
	`repvgg_a2`	78.4 ± 0.10	0.527 ± 0.006	0.457 ± 0.005	0.350 ± 0.004	0.239 ± 0.004	0.207 ± 0.003
	Pearson coefficient (r)		-0.05	0.01	-0.87	-0.14	0.61
RV	`repvgg_b0`	79.8 ± 0.22	0.568 ± 0.021	0.516 ± 0.017	0.435 ± 0.014	0.344 ± 0.005	0.317 ± 0.002
	`repvgg_b1`	78.8 ± 0.19	0.574 ± 0.002	0.525 ± 0.001	0.449 ± 0.001	0.354 ± 0.001	0.320 ± 0.000
	`repvgg_b2`	79.0 ± 0.10	0.573 ± 0.003	0.526 ± 0.003	0.446 ± 0.002	0.347 ± 0.004	0.312 ± 0.004
	Pearson coefficient (r)		-1.00	-0.97	-1.00	-0.85	-0.07
SN	`shufflenetv2_x0_5`	81.8 ± 0.42	0.453 ± 0.018	0.365 ± 0.014	0.246 ± 0.010	0.144 ± 0.007	0.117 ± 0.005
	`shufflenetv2_x1_0`	78.4 ± 0.35	0.475 ± 0.005	0.395 ± 0.006	0.282 ± 0.003	0.164 ± 0.006	0.131 ± 0.006
	`shufflenetv2_x1_5`	77.5 ± 0.42	0.478 ± 0.011	0.396 ± 0.009	0.282 ± 0.009	0.167 ± 0.006	0.135 ± 0.006
	`shufflenetv2_x2_0`	77.0 ± 0.43	0.495 ± 0.010	0.417 ± 0.010	0.301 ± 0.008	0.183 ± 0.007	0.150 ± 0.006
	Pearson coefficient (r)		-0.94	-0.95	-0.97	-0.94	-0.91
			CIFAR-100				
RV	`repvgg_a0`	87.3 ± 0.12	0.298 ± 0.003	0.227 ± 0.003	0.146 ± 0.005	0.085 ± 0.004	0.070 ± 0.003
	`repvgg_a1`	86.7 ± 0.08	0.310 ± 0.004	0.241 ± 0.007	0.161 ± 0.003	0.095 ± 0.002	0.078 ± 0.001
	`repvgg_a2`	85.3 ± 0.42	0.314 ± 0.001	0.251 ± 0.002	0.171 ± 0.002	0.103 ± 0.002	0.086 ± 0.002
	Pearson coefficient (r)		-0.86	-0.94	-0.93	-0.96	-0.98
RV	`repvgg_b0`	82.1 ± 0.25	0.355 ± 0.004	0.306 ± 0.005	0.227 ± 0.006	0.171 ± 0.005	0.151 ± 0.004
	`repvgg_b1`	87.4 ± 0.20	0.345 ± 0.007	0.266 ± 0.009	0.210 ± 0.010	0.140 ± 0.005	0.118 ± 0.003
	`repvgg_b2`	87.1 ± 0.32	0.324 ± 0.002	0.284 ± 0.002	0.204 ± 0.000	0.155 ± 0.002	0.147 ± 0.001
	Pearson coefficient (r)		-0.71	-0.92	-0.95	-0.89	-0.63
SN	`shufflenetv2_x0_5`	88.1 ± 0.13	0.259 ± 0.018	0.188 ± 0.015	0.114 ± 0.010	0.061 ± 0.006	0.048 ± 0.005
	`shufflenetv2_x1_0`	87.6 ± 0.37	0.281 ± 0.010	0.207 ± 0.007	0.127 ± 0.004	0.072 ± 0.001	0.057 ± 0.001
	`shufflenetv2_x1_5`	87.0 ± 0.05	0.273 ± 0.007	0.206 ± 0.002	0.128 ± 0.005	0.072 ± 0.003	0.058 ± 0.002
	`shufflenetv2_x2_0`	86.0 ± 0.03	0.294 ± 0.012	0.228 ± 0.010	0.150 ± 0.008	0.086 ± 0.005	0.072 ± 0.005
	Pearson coefficient (r)		-0.87	-0.95	-0.97	-0.97	-0.97

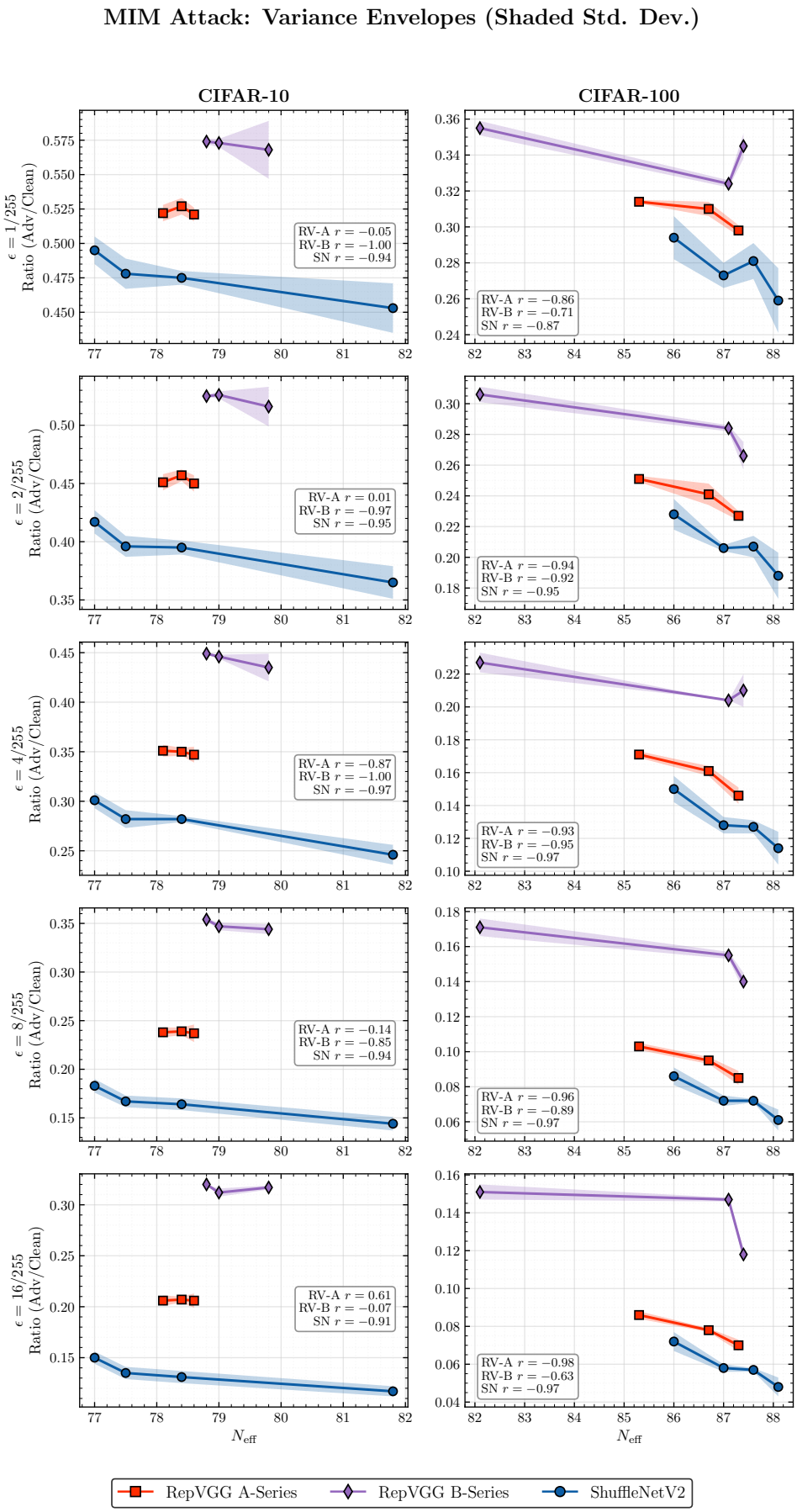

Figure 1: Robustness Ratios for the MIM Attack across budgets ϵ.

Square Attack

Table 2: Mean Robustness Ratios ($Adv/Clean$) ± std for the Square Attack across budgets ϵ. Architecture families: **ShuffleNetV2** (SN) and **RepVGG** (RV). r denotes the Pearson correlation coefficient.

Fam.	Model	$N_{eff} \downarrow$	$\epsilon = \frac{1}{255}$	$\epsilon = \frac{2}{255}$	$\epsilon = \frac{4}{255}$	$\epsilon = \frac{8}{255}$	$\epsilon = \frac{16}{255}$
			CIFAR-10				
RV	`repvgg_a0`	78.1 ± 0.12	0.558 ± 0.006	0.546 ± 0.007	0.522 ± 0.005	0.471 ± 0.006	0.377 ± 0.008
	`repvgg_a1`	78.6 ± 0.15	0.558 ± 0.002	0.545 ± 0.002	0.520 ± 0.002	0.470 ± 0.005	0.376 ± 0.010
	`repvgg_a2`	78.4 ± 0.10	0.565 ± 0.005	0.553 ± 0.005	0.528 ± 0.007	0.478 ± 0.007	0.380 ± 0.006
	Pearson coefficient (r)		0.15	0.00	-0.10	-0.08	-0.20
RV	`repvgg_b0`	79.8 ± 0.22	0.544 ± 0.022	0.535 ± 0.023	0.516 ± 0.021	0.476 ± 0.017	0.399 ± 0.013
	`repvgg_b1`	78.8 ± 0.19	0.554 ± 0.003	0.544 ± 0.003	0.526 ± 0.002	0.484 ± 0.001	0.410 ± 0.000
	`repvgg_b2`	79.0 ± 0.10	0.552 ± 0.002	0.542 ± 0.003	0.522 ± 0.004	0.487 ± 0.004	0.410 ± 0.001
	Pearson coefficient (r)		-1.00	-1.00	-0.98	-0.90	-0.98
SN	`shufflenetv2_x0_5`	81.8 ± 0.42	0.503 ± 0.017	0.486 ± 0.017	0.453 ± 0.018	0.390 ± 0.018	0.287 ± 0.018
	`shufflenetv2_x1_0`	78.4 ± 0.35	0.522 ± 0.008	0.507 ± 0.006	0.476 ± 0.004	0.419 ± 0.005	0.316 ± 0.006
	`shufflenetv2_x1_5`	77.5 ± 0.42	0.524 ± 0.011	0.508 ± 0.011	0.479 ± 0.011	0.419 ± 0.009	0.315 ± 0.005
	`shufflenetv2_x2_0`	77.0 ± 0.43	0.538 ± 0.009	0.523 ± 0.009	0.492 ± 0.007	0.434 ± 0.007	0.329 ± 0.005
	Pearson coefficient (r)		-0.96	-0.95	-0.97	-0.97	-0.97
			CIFAR-100				
RV	`repvgg_a0`	87.3 ± 0.12	0.352 ± 0.006	0.337 ± 0.005	0.306 ± 0.005	0.250 ± 0.005	0.158 ± 0.004
	`repvgg_a1`	86.7 ± 0.08	0.359 ± 0.002	0.345 ± 0.003	0.317 ± 0.004	0.261 ± 0.005	0.169 ± 0.005
	`repvgg_a2`	85.3 ± 0.42	0.359 ± 0.006	0.345 ± 0.004	0.318 ± 0.003	0.266 ± 0.003	0.176 ± 0.001
	Pearson coefficient (r)		-0.73	-0.77	-0.81	-0.90	-0.93
RV	`repvgg_b0`	82.1 ± 0.25	0.374 ± 0.002	0.343 ± 0.001	0.231 ± 0.003	0.143 ± 0.002	0.110 ± 0.001
	`repvgg_b1`	87.4 ± 0.20	0.325 ± 0.003	0.314 ± 0.002	0.225 ± 0.004	0.135 ± 0.004	0.070 ± 0.003
	`repvgg_b2`	87.1 ± 0.32	0.366 ± 0.002	0.284 ± 0.002	0.224 ± 0.005	0.134 ± 0.004	0.108 ± 0.003
	Pearson coefficient (r)		-0.81	-0.74	-0.87	-0.97	-0.71
SN	`shufflenetv2_x0_5`	88.1 ± 0.13	0.305 ± 0.011	0.288 ± 0.013	0.258 ± 0.014	0.205 ± 0.014	0.132 ± 0.010
	`shufflenetv2_x1_0`	87.6 ± 0.37	0.327 ± 0.012	0.313 ± 0.011	0.283 ± 0.010	0.227 ± 0.004	0.143 ± 0.002
	`shufflenetv2_x1_5`	87.0 ± 0.05	0.320 ± 0.009	0.305 ± 0.010	0.278 ± 0.009	0.225 ± 0.005	0.145 ± 0.005
	`shufflenetv2_x2_0`	86.0 ± 0.03	0.336 ± 0.011	0.323 ± 0.010	0.295 ± 0.010	0.244 ± 0.011	0.162 ± 0.009
	Pearson coefficient (r)		-0.84	-0.83	-0.88	-0.93	-0.97

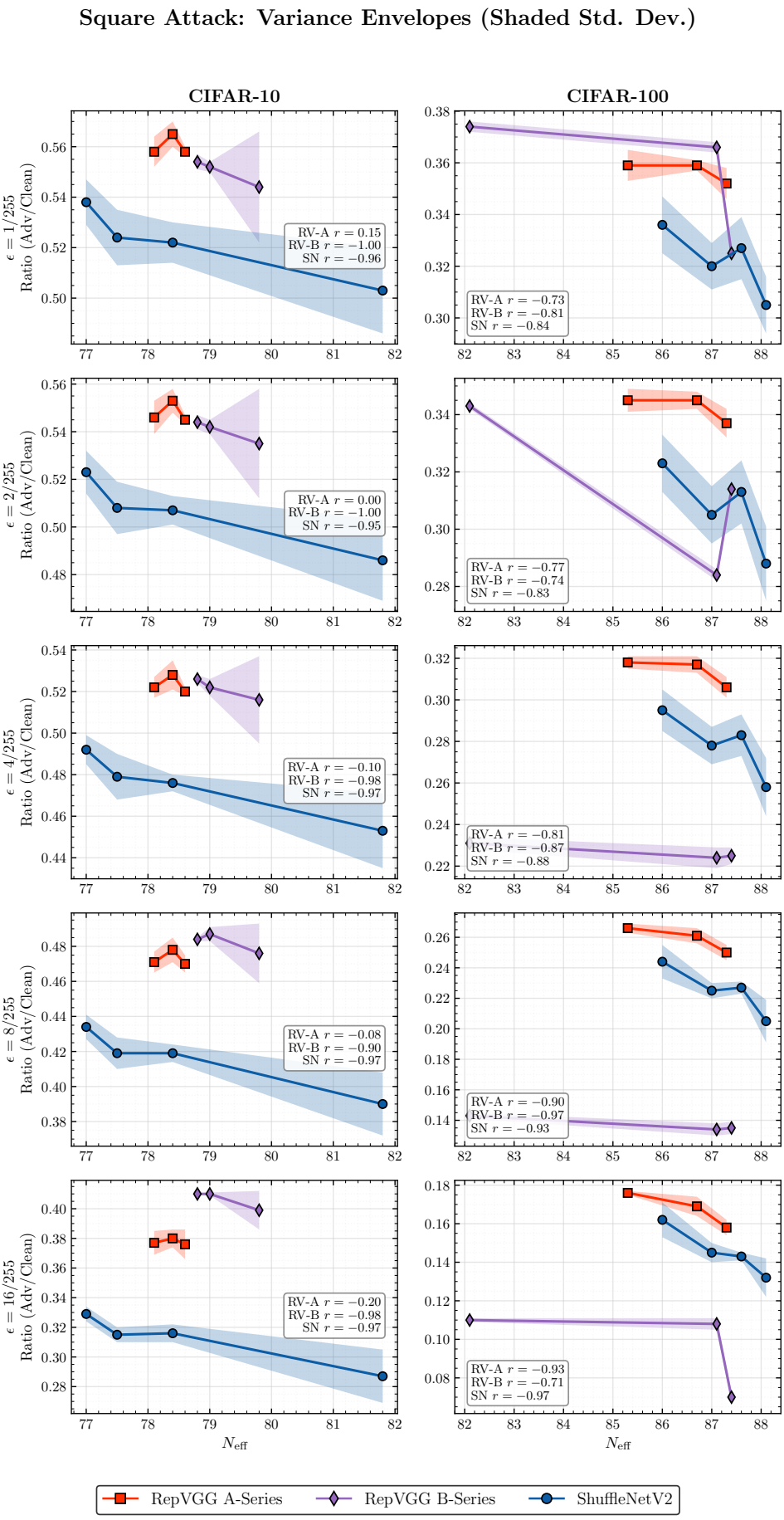

Figure 2: Robustness Ratios for the Square Attack across budgets ϵ.